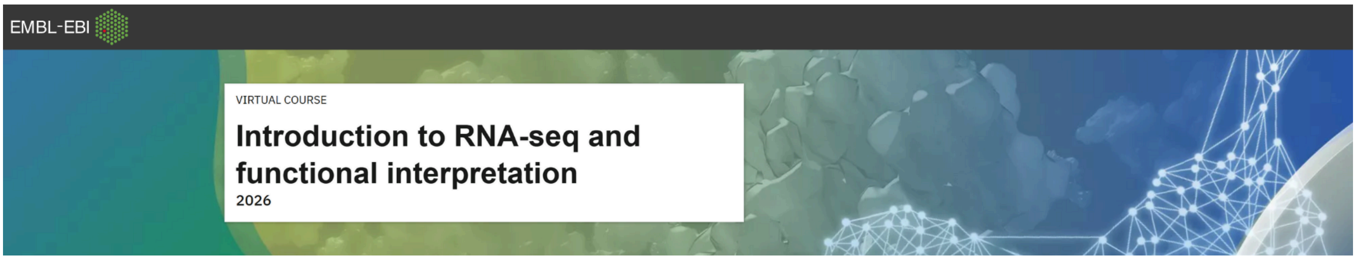

Tuesday, February 10th, 2026

Preview of the session *"Long-read RNA-seq analysis"*

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During the EMBL-EBI main event of the course "Introduction to RNA-seq and functional interpretation", we will share a session about analyzing data from long-read RNA sequencing. The session will happen on day 3 (February 11th, 2026) from 13:00 to 17:00 (UK Time). Check the complete program for more details.

We expect not only introduce you to long-read RNA-seq technologies, but also provide you a clear overall glimpse of how to transform noisy data into valuable and comprehensive results that provide biologically relevant information. To be very brief, the competencies aimed for you to develop after the session are:

> 1. Understand how long-read RNA-seq (lrRNA-seq) technologies work and differ between each other, also comparing them to short-read approaches, and highlighting their advantages, potentials and limitations.
> 2. Learn the main general steps of lrRNA-seq analysis and comprehend the raw results generated
> 3. Learn how to engineer raw noisy data into organized biological metadata for further exploratory analysis

Before starting the course, we are recommending you some materials, tutorials and reading suggestions so that you can prepare. This is specially important if you are a newbie to coding, IDEs[1], bioinformatics and RNA-seq in general. You don't need to be an expert, because we are going to use an already built and defined workflow. However, we expect you to be able to minimally understand what an algorithm is doing. Therefore, we will provide you the main important materials to get familiar with some concepts and tools before the course begins. Additionally, if you wish to further study about lrRNA-seq, a document with complementary material recommendations (non-mandatory for the course) is also attached.

Pre-course instructions

Bioinformatics resources to be used

Linux-based command-line and R tutorials suggested by the event's organization team

As the event's organization explained at the official website:

> *"Some experience with R and the linux-based command line is beneficial, but not essential. During the course some of the practicals will make use of a Linux-based command line interface, and R statistical packages. We recommend completing some basic tutorials on this topic in preparation for the upcoming course. […] Regardless of your current knowledge, we encourage successful participants to use these, and other materials, to prepare for attending the course and future work in this area."*

There are many tutorials available online. These are the free tutorials recommended:

1. Introduction to the Unix environment – https://swcarpentry.github.io/shell-novice/index.html

2. Introduction to R – https://swcarpentry.github.io/r-novice-inflammation/

Materials we suggest for you to read

It is also important that you check about the tidyverse collection package[1] – which is really useful and will play the most important role – noisy data engineering – from the R scripts we have built for the course. A very good reference to introduce yourself to R and tidyverse at the same time is the 2nd edition of the book "R for data science: Import, Tidy, Transform, Visualize, and Model Data"[2] written by Hadley Wickham, Mine Çetinkaya-Rundel & Garrett Grolemund (read it online for free here). Other additional papers and learning materials are mentioned in the complementary recommendations document.

The main bioinformatics tools we are going to use

We will implement a step-by-step core workflow – similar to some parts of the LongNonCoder nextflow pipeline (under development at LIB & BioME), and the nf-core/nanoseq pipeline[3], but excluding the discovery of novel transcripts. This workflow focuses on the main essential long-read data pre-processing, followed by a full reference transcriptome characterization and other downstream analysis for differential isoform expression and functional enrichment.

You can see evaluations and validations for many of these and other tools on some of the latest benchmarkings[3–5], and also this review paper[6]. Furthermore, if you wish to do so, take a look at the main bioinformatics tools we are going to use to run our workflow:

Table 1: Order of the main tools used in the workflow

Step	Tool	Key Functionality
1, 3	NanoComp	Compares multiple runs of long-read sequencing data in FASTQ or BAM formats. It generates plots to visualize differences in read length, quality scores, and yield across samples.[7]
2	Chopper	A tool for trimming and filtering long reads. It is used to remove low-quality reads or reads shorter/longer than a specific length to improve downstream analysis.[7]

Step	Tool	Key Functionality
4	Minimap2	A versatile, high-performance aligner. It is the “gold standard” for mapping long reads (DNA or mRNA/cDNA) to a reference genome or transcriptome.[8]
5	Samtools	It processes SAM/BAM/CRAM files (sorting, indexing, and converting) so they can be read by other tools.[9]
6	Bambu	An R package used for multi-sample transcript discovery and quantification using long-read RNA-seq data. It helps identify novel isoforms.[10]
7	MultiQC	Aggregates log files and results from many different bioinformatics tools (like NanoComp and Samtools) into a single, interactive HTML report.[11]
8	biomaRt	Converts different IDs (e.g., Ensembl Gene ID to Gene Symbol) and retrieves additional annotations or sequences from databases.[12]
9	DESeq2	Statistically compares gene/transcript counts between conditions (e.g., treated vs. control) to find what changed significantly.[13]
10	Enrichr / g:Profiler	Takes a list of genes and finds annotations, such as biological pathways, functions and cell components they are significantly associated with.[14,15]

Valuable tip

Check the complementary material recommendations document for more details!

Google Colaboratory (Colab)

Google Colab (short for **Colaboratory**) is a cloud-based service provided by Google that allows you to write and execute code – primarily **Python**, but also works with **R** – entirely through your web browser. Built on top of the**Jupyter Notebook** environment, it provides an interactive space where you can combine executable code, rich text (Markdown), and visualizations in a single document. For bioinformaticians, Colab has become a transformative tool because it bypasses many of the traditional barriers to high-performance computational biology.

Bioinformatics often involves processing massive datasets (like genomic sequences) and running complex algorithms that require significant computing power. Colab addresses these challenges in several ways:

1. **Free Access to High-Performance Hardware**
   Bioinformatics tasks, especially those involving deep learning (like **AlphaFold** for protein structure prediction) or large-scale sequence alignments, require powerful **GPUs** (Graphics Processing Units) and **TPUs** (Tensor Processing Units). Colab provides free access to these resources, which would otherwise be very expensive to purchase or rent via standard cloud providers.

2. **Seamless Collaboration and Reproducibility**
   Scientific research relies on reproducibility. Since Colab notebooks are stored in Google Drive and can be linked to GitHub, you can share a live, executable version of your analysis with a single link. Reviewers or collaborators can run your exact code, on the same hardware, without having to set up a local environment

3. **Integration with Big Data**
   Bioinformatics increasingly relies on cloud storage. Colab integrates natively with:

   - **Google Drive:** For storing large FASTQ or BAM files.
   - **Google Cloud Storage:** For accessing public datasets like those from the 1000 Genomes Project.
   - **GitHub:** For version control and sharing pipelines.

Here are some videos and reading materials about Google Colab that we recommend:

1. Welcome to Colab
2. Get started with Google Colaboratory
3. Google Colab features you may have missed

Final schedule

The session is planned to be fairly balanced between theory and practical training. A few days before the event, you will be sent the final instructions, all materials of our presentations and practical training. For now, you can check the main schedule below:

Table 2: Long-read RNA-seq analysis session schedule

UK Time	Theme
13:00 - 13:50	**An overview about long-read RNA-seq** 1. Discuss what is long-read RNA sequencing (Oxford Nanopore and PacBio) 2. Evolution and differences from short-read technologies 3. Main steps of data analysis – dealing with noise 4. Importance of isoform-level analysis 5. Advantages, potential, and limitations
13:50 - 14:00	Discussion / Questions & Answers (1)
14:00 - 14:20	Break
14:20 - 14:55	**Preparing for practical training** 1. Memory refresh – recapitulating key concepts 2. Detailed step-by-step analysis workflow 3. Presentation of practical training material
14:55 - 15:00	Discussion / Questions & Answers (2)
15:00 - 16:30	**Practical training** 1. QC and Mapping 2. Isoform-level transcriptome assembly and quantification 3. Differential gene and transcript expression analysis 4. Transcriptome characterization 5. Functional enrichment analysis based on gene to isoform-level relationships
16:30 - 16:45	Conclusive overview and discussion of the results for biological interpretation

UK Time	Theme
16:45 - 17:00	1. Discussion / Questions & Answers (3) 2. Final considerations and ending of the session

Final considerations

Finally, it is crucial to understand that the course depends mostly on attention and teamwork (between trainers and participants). We expect you to engage by asking/answering questions and getting involved in the open discussions. Furthermore, you are also welcome to leave any message in the chat as well – there, you can make questions, comments, tell us about a related experience you had, and even help another participant. Remember that we are learning from you too. We will do our best to listen and assist everyone, giving you all the opportunity to fully participate in this shared learning experience.

> **Last important note**
>
> If you have any doubts or require any special accessibility assistance about this document or the course, do not hesitate to contact us.

> **Contact both of us through e-mail**
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Se you all very soon 😊

Best regards,

Vinícius Maracajá-Coutinho and Bárbara Borges

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Footnotes

1. Integrated Development Environment – a software application that provides a comprehensive suite of tools for software developers to write, edit, test, and debug code efficiently within a single graphical user interface (GUI). It acts as a "one-stop shop" that consolidates various development utilities, eliminating the need to use and configure separate applications for coding, compiling, and testing ↩